

A Counterexample to Directed Unions for Algebraic Envelopes

Run `fa_banach_001`, agent `agent_lane_04`

June 14, 2026

Status

Claimed counterexample. The construction below gives a negative answer to the question in Ferenczi–Lopez-Abad, *Envelopes in Banach spaces*, arXiv:2112.11795, asking whether the directed-union formula for isometric envelopes remains true for algebraic envelopes.

Source Problem

The source paper proves the following directed-union formula for the isometric envelope.

group in separable reflexive Banach spaces [Lan93], we have the following.

Corollary 3.3. *Assume X is reflexive and either separable or LUR. Let $Y \subseteq X$ be the closure of a directed sequence $(Y_i)_{i \in I}$ of subspaces of X . Then*

$$\text{Env}(Y) = \overline{\bigcup_{i \in I} \text{Env}(Y_i)}.$$

Immediately after it, the authors ask whether the analogous statement holds for algebraic envelopes.

20

V. FERENCZI, J. LOPEZ-ABAD

We do not know if the above is true for the algebraic envelope.

3.1. Envelopes in rearrangement invariant spaces.

In the notation of the source paper, the algebraic envelope of a subspace $Y \subseteq X$ is

$$\text{Env}^{\text{alg}}(Y) = \text{Fix}(\text{Stab}_{\text{Isom}(X)}(Y)),$$

the subspace of all vectors fixed by every surjective linear isometry of X which fixes Y pointwise.

Thus the question is whether, when X is reflexive and separable or LUR and $Y = \overline{\bigcup_i Y_i}$ for a directed family of subspaces,

$$\text{Env}^{\text{alg}}(Y) = \overline{\bigcup_i \text{Env}^{\text{alg}}(Y_i)}$$

must hold.

Idea of the Proof

Finite-dimensional sublattices of L_p have no hidden algebraic envelope: inside each atom of the finite partition, measure-preserving isometries can move equal-measure pieces around, so a fixed function must be constant on each atom.

The limit sublattice below is different. It remembers the first coordinate of $[0, 1] \times \{0, 1\}$, but not the two-point fiber. Because the two fiber weights are unequal, any measure automorphism that fixes every base set must also fix each sheet of the two-point fiber. Therefore the indicator of one sheet is fixed by all isometries stabilizing the limit subspace, even though it is not itself base-measurable. This creates a strict algebraic envelope at the limit while all finite-stage envelopes stay trivial.

Counterexample

Theorem 1. *Let $1 < p < \infty$, $p \neq 2$. There are a separable reflexive space X and an increasing sequence of subspaces $Y_n \subseteq X$, with*

$$Y = \overline{\bigcup_{n=1}^{\infty} Y_n},$$

such that

$$\overline{\bigcup_{n=1}^{\infty} \text{Env}^{\text{alg}}(Y_n)} = Y \quad \text{but} \quad \text{Env}^{\text{alg}}(Y) \supsetneq Y.$$

Consequently the directed-union formula for algebraic envelopes is false.

Proof. Let

$$\Omega = [0, 1] \times \{0, 1\}, \quad \mu = \lambda \otimes \left(\frac{1}{3}\delta_0 + \frac{2}{3}\delta_1 \right),$$

where λ is Lebesgue measure. Put $X = L_p(\Omega, \mu)$. This is a real separable reflexive L_p -space. Since Ω is a standard nonatomic probability space, X is isometric to the usual $L_p[0, 1]$.

Let $\mathcal{A}$ be the base sigma-algebra

$$\mathcal{A} = \{E \times \{0, 1\} : E \in \mathcal{B}([0, 1])\},$$

and let $Y = L_p(\mathcal{A})$. For $n \geq 1$, let $\mathcal{A}_n$ be the finite sigma-algebra generated by the dyadic intervals of length 2^{-n} in the first coordinate, again with the whole two-point fiber attached. Set

$$Y_n = L_p(\mathcal{A}_n).$$

Then Y_n is increasing and $\overline{\bigcup_n Y_n} = Y$.

We first show that $\text{Env}^{\text{alg}}(Y_n) = Y_n$. Let $f \notin Y_n$. Then on some atom C of $\mathcal{A}_n$, the restriction of f is not essentially constant. Hence there are real numbers $a > b$ and positive-measure sets $B, C' \subseteq C$ of equal measure such that $f \geq a$ on B and $f \leq b$ on C' . As C is nonatomic, there is a measure-preserving automorphism of Ω that swaps B and C' , is the identity outside small pieces of the atom C , and preserves every atom of $\mathcal{A}_n$. The induced composition isometry of $L_p(\Omega)$ fixes Y_n pointwise but does not fix f . Therefore no $f \notin Y_n$ lies in $\text{Env}^{\text{alg}}(Y_n)$, and $\text{Env}^{\text{alg}}(Y_n) = Y_n$.

It follows that

$$\overline{\bigcup_n \text{Env}^{\text{alg}}(Y_n)} = \overline{\bigcup_n Y_n} = Y.$$

Now let

$$A_0 = [0, 1] \times \{0\}.$$

We claim that $\mathbf{1}_{A_0} \in \text{Env}^{\text{alg}}(Y)$. Let T be a surjective linear isometry of $L_p(\Omega, \mu)$ fixing Y pointwise. Since $\mathbf{1}_\Omega \in Y$, $T\mathbf{1}_\Omega = \mathbf{1}_\Omega$. By the Banach–Lamperti description of L_p -isometries for $p \neq 2$, T is therefore induced by a measure-algebra automorphism θ of Ω . Because T fixes $Y = L_p(\mathcal{A})$ pointwise, θ fixes every base set $E \times \{0, 1\}$.

Let $B = \theta(A_0)$. For every Borel $E \subseteq [0, 1]$,

$$\mu(B \cap (E \times \{0, 1\})) = \mu(A_0 \cap (E \times \{0, 1\})) = \frac{1}{3}\lambda(E).$$

Disintegrate over the first coordinate. For a.e. x , the fiber $B_x \subseteq \{0, 1\}$ has conditional measure in $\{0, \frac{1}{3}, \frac{2}{3}, 1\}$. The preceding identity for all E says that this conditional measure is $\frac{1}{3}$ a.e. Hence $B_x = \{0\}$ for a.e. x , and so $B = A_0$ in the measure algebra. Thus $T\mathbf{1}_{A_0} = \mathbf{1}_{A_0}$.

Since this holds for every surjective isometry fixing Y pointwise, $\mathbf{1}_{A_0} \in \text{Env}^{\text{alg}}(Y)$. But $\mathbf{1}_{A_0} \notin Y$, because it is not measurable with respect to the base sigma-algebra $\mathcal{A}$. Therefore $\text{Env}^{\text{alg}}(Y) \supsetneq Y$, as required. $\square$

Verification Notes

The proof uses no computation. The finite-stage step uses only the nonatomicity of each atom of the finite base partition. The limit-stage strictness uses the unequal conditional weights $1/3$ and $2/3$, which make the sheet A_0 recognizable among all measurable subsets after conditioning on the base.

The Banach–Lamperti theorem is used only in the standard form that a surjective linear isometry of L_p , $p \neq 2$, is a weighted composition operator; since the isometry fixes the constant 1, the weight is 1.

Novelty Check

Before promotion, I searched the run indexes for arXiv ‘2112.11795’, envelope, algebraic envelope, fixed-point subalgebra, and directed-union terms. The only hits were the prior unfinished attempt for this same route and unrelated uses of the words ‘envelope’ or ‘centralizer’.

I also searched the web for exact and close phrases including:

- "We do not know if the above is true for the algebraic envelope",
- "algebraic envelope" "directed" "union" "Banach",
- "Env^{alg}" "directed" "union",
- "Env^{alg}" "L_p" "fixed-point subalgebra".

No later paper or explicit answer to this question was found in that bounded search. The counterexample is close to the source paper’s own fixed-point subalgebra obstruction, but the source does not instantiate the unequal-fiber example above.

Limitations

This packet disproves the directed-union formula for algebraic envelopes. It does not address whether additional assumptions, such as WOT-closedness of the relevant stabilizer or a fixed-point hypothesis on the limiting sigma-algebra, recover a positive theorem.

Human Review Recommendation

Review the Banach–Lamperti reduction and the conditional-fiber calculation showing that a base-fixing measure automorphism must fix A_0 . These are the only non-formal points in the counterexample.

References

- [1] V. Ferenczi and J. Lopez-Abad, *Envelopes in Banach spaces*, arXiv:2112.11795.
- [2] J. Lamperti, *On the isometries of certain function-spaces*, Pacific J. Math. **8** (1958), 459–466.